VANSH GUPTA

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PROFESSIONAL SUMMARY

Computer Science graduate specializing in production-grade GenAI systems — multi-agent RAG pipelines, LLM reasoning agents, and document-intelligence tools built with LangChain and LangGraph.

PROJECTS

V-RAG — Verified Retrieval-Augmented Generation *Aug 2026*

Python, LangChain, LangGraph, LangSmith, OpenAI, Chroma, Pydantic

- **Multi-agent architecture:** designed a 3-agent system (Responder, Prosecutor, Judge) for evidence-grounded QA over a legal document corpus, detecting unsupported claims and iteratively revising responses.
- **Evaluation:** built a 50-question evaluation suite spanning fact retrieval, temporal reasoning, and contradiction-handling; achieved 0.745 case-level recall and 0.951 MRR at k=7, with hit rate reaching 1.00 for case/citation-level retrieval.
- **Evaluation:** Ran LLM-as-judge evaluation across correctness, faithfulness, and groundedness; achieved 82% correctness and 95.4% average faithfulness with only 6% hallucination rate on the baseline answer set.

Compliance Agent — Intelligent Document Compliance & Automation *May 2025*

Python, LangChain, Gemini API, PostgreSQL, OCR, PDF Parsing

- **Document reasoning agent:** built an AI agent that reasons across structured and unstructured documents to automate regulatory compliance checks, replacing manual review.
- **Ingestion pipeline:** implemented OCR (Google Vision) and PDF parsing to extract structured data from unstructured compliance documents.
- **CI/CD:** Set up a GitHub Actions pipeline separating deterministic unit tests from live-API integration tests, with a dedicated workflow to regenerate JSON/Markdown eval reports as build artifacts on demand.

HOPE — AI-Powered Mental Health Platform *2022 – 2023*

HTML, CSS, JavaScript, OpenAI, Whisper API

- **Conversational AI:** full-stack CBT therapy platform using GPT + Whisper APIs for voice/text conversations; reached 60+ monthly active users and was backed by Microsoft for Startups (GDSC Solutions Challenge).

EXPERIENCE

Software Developer Intern | Aseries Envirotek India Pvt. Ltd. — Remote *Feb 2025 – Sep 2025*

- Built a React.js / Node.js / PostgreSQL portal automating Environmental Impact Assessment (EIA) report generation, replacing manual drafting.
- Engineered dynamic document generation with placeholder replacement across official forms (Form 1, Form 1A, Conceptual Plan, EMP) while preserving original formatting for PDF/DOC outputs.
- Built an admin dashboard that reduced manual documentation effort and improved client-submission tracking; owned architecture, backend, and frontend independently.

Freelance Developer | Self-employed — Remote *Sep 2025 – Present*

- Built MMABOXING, a Shopify storefront for a combat-sports brand, translating business requirements into a production-ready online store.
- Developed Flasseq, a full-stack clothing e-commerce platform using EverShop (React theme) with Razorpay payment integration.

Frontend Developer Intern | Trajectorie — Remote *Sep 2023 – Jul 2024*

- Led frontend development on a React.js quizzing platform, improving responsiveness and cross-browser reliability via integrated RESTful APIs.

Software Developer Intern | Digiproctor — Remote *Jun 2023 – Aug 2023*

- Built an internal interviewing platform (JavaScript/Node.js) with an integrated Rasa chatbot and a browser-based PDF editor, reducing query resolution time and boosting engagement.

SKILLS

ML / GenAI: GenAI, LLMs, Prompt Engineering, LangChain, LangGraph, LangSmith, RAG, Vector Databases, MCP, LLM Evals, LLM Gateways, Gaurdrails, PyTorch, Scikit-learn, Pandas, FastAPI

Web & Programming: Python, JavaScript, SQL, React.js, Node.js, REST APIs, HTML5, CSS, MySQL, PostgreSQL, Git, CI/CD

EDUCATION

B.E. Computer Science | Birla Institute of Technology and Science, Pilani — Dubai, UAE *2021 – 2025*

LEADERSHIP & AWARDS

- **Google Developer Student Club — Technical Executive** (Aug 2023 – Jun 2024): led the Web Development network, mentored 60+ students in HTML/CSS/JS workshops, and organized CTFs engaging 300+ participants.
- **Awards:** NASA Space Apps Challenge — UAE Winner & Global Nominee (2023); Emirati Sign Language Project — 2nd Place; ACM-CIIED Ideathon — Winner; RIFT CTF — 3rd; Exploit3rs CTF (UAE) — 5th/120.